

PisoFire

High-Demand Sneaker E-commerce Platform

Group D

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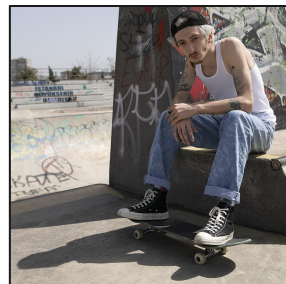
What is PisoFire?

- **Product** : Cloud-native e-commerce platform dedicated to premium sneaker releases.
- **Industry Domain** : High-volume retail focusing on "Flash Sales" .
- **SRE Challenge**: The system must operate reliably during periods of constant traffic; maintain availability and data integrity during “Flash Sales”.

Critical Path

Flash Sales Buying Path

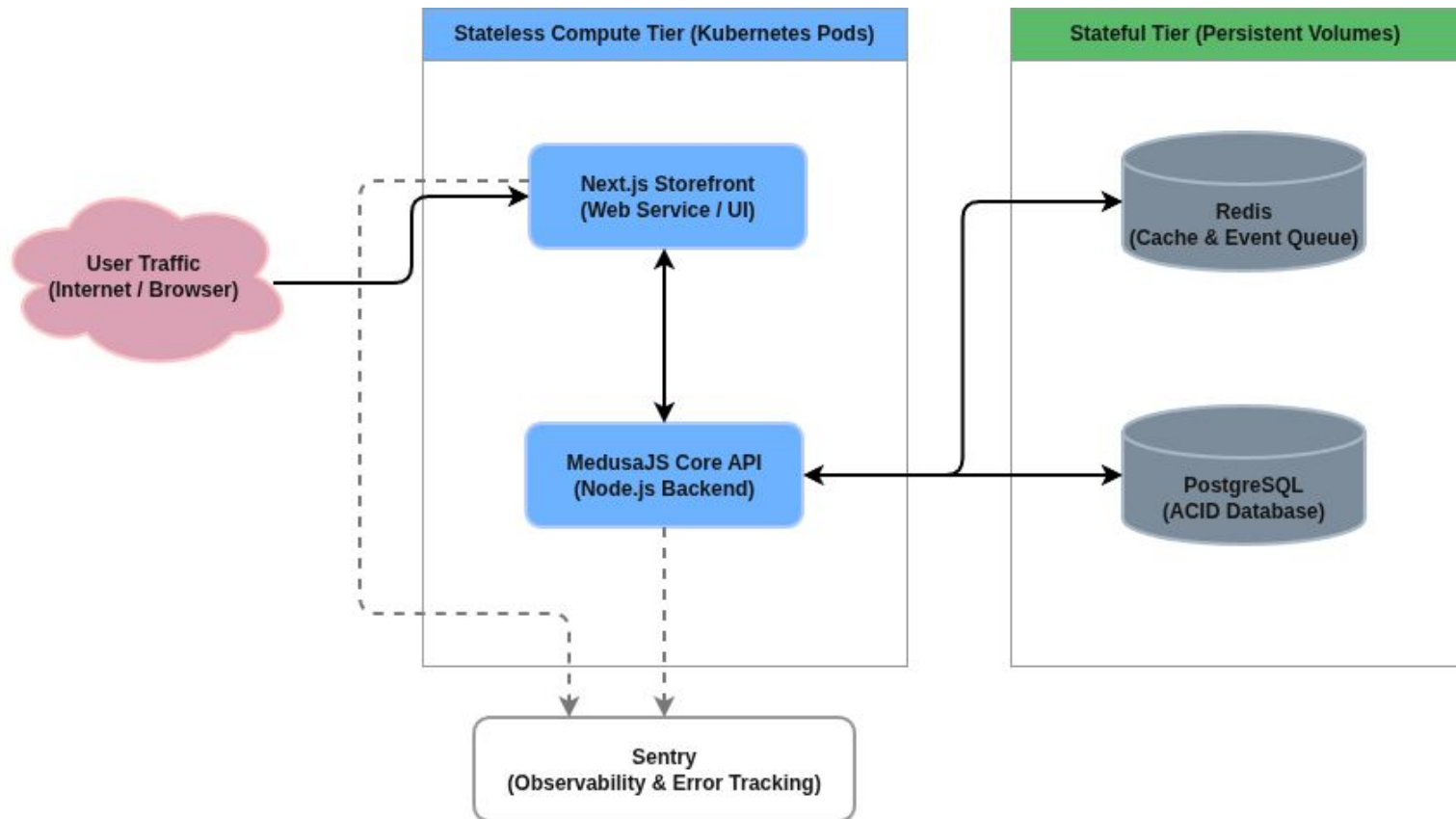
- User goes to flash sale landing page after launching time
- User can view the product's details
- User selects a size
- User adds item to cart
- User goes to checkout
- Payment is processed
- Inventory is decreased



TECHNICAL PLANNING



Architecture



Why MedusaJS?



Scalability

E-commerce framework made with scalability in mind.



Modules Pre-Integration

Medusa already has the Database, Backend and Frontend pre-integrated.



Caching with Redis

Module for caching and eventBus with Redis out of the box.

“Operations First” Strategy



Data Seeding



Direct API for Orders



IaC



Observability

Operational Foresight

Critical Dependencies



Primary Database



Redis

“Hardest” Components



Database Scaling



Redis Recovery

Operational Foresight

“Can you do updates to each component?”

- **Backend:** Updates instances one by one. The load balancer sends traffic to the "old" instances while the "new" ones boot up.
- **PostgreSQL Database:** Backup and then update with database migrations. (Expand and contract)
- **Redis:** Since we will use a Redis Cluster, we can update the software on one node at a time without the entire cache going offline.

Thanks for your attention!

Any feedback?

